# Daily Drilling Report
## SPD19B-03
## 4/10/2014

PETROPARS Oil & Gas Developer

N.I.O.C. Pars Oil & Gas Company

### Well Data

| Well Name | Operator | Field Name | Well Type 1 | Rig | Code |
|---|---|---|---|---|---|
| SPD19B-03 | Petropars Ltd. | South Pars Gas Field | Development | Deep Driller, DD6 | SP19-3B-01-DL-FOR-0... |
| **Report Number** | **Hole Section** | **Days LTI (days)** | **Water Depth (m)** | **Original KB Elevation (m)** | **Head Count** |
| 16.0 | 16" | 37.00 | 72.83 | 31.70 | 98.0 |
| **Last Cas** | **Daily Cost Total** | **Daily Mud Cost** | **Cum Cost To Date** | **AFE Number** | **Cum Time Log Days (days)** |
| 18 5/8" Shoe, 870.00mKB | 204,530.15 | 10,061.48 | 3,735,020.05 | | 13.85 |
| **Depth Start (mKB)** | **Depth End (mKB)** | **Depth End (TVD) (mKB)** | **Depth Progress (m)** | **Drilling Hours (hrs)** | **Average ROP (m/hr)** |
| 960.00 | 1,184.00 | 1,133.35 | 224.00 | 18.20 | 12.3 |

**Operations at Report Time**
Drilling @ 1240m.

**Operations Summary**
Directional drilling 16" hole section f/ 960m to 1184m.

**Operations next 24hrs**
Cont' drilling 16" hole.

### Supervisors Contact

| Job Contact | Position |
|---|---|
| Mojtaba Parvaresh | Drilling Supt. |
| Mohammad Ali Rezaei | Senior Drilling Sup. |
| Vahab Kari | Night Drilling Sup. |
| Hamed Daneshkhah | Drilling Engineer |
| Yaghub Mohammadi | Geologist |
| GH. R. Babaei | POGC Rep. |

### HSE Data

| Type | Last Date | Next Date |
|---|---|---|
| Abandon Drill | 3/9/2014 | 4/29/2014 |
| BOP Test | 4/7/2014 | 4/29/2014 |
| Chemical Spill Drill | 3/29/2014 | 5/29/2014 |
| Fire Drill | 3/9/2014 | 4/29/2014 |
| H2S Drill | 4/4/2014 | 5/5/2014 |
| Heli Crash Drill | 3/22/2014 | 5/22/2014 |
| Medical Drill | 4/4/2014 | 6/4/2014 |
| MOB Drill | 2/28/2014 | 4/30/2014 |
| Pit Drill | 4/9/2014 | 5/25/2014 |
| Trip Drill | 4/8/2014 | 5/29/2014 |

### STOP Cards

**Number of Reports:** 33

### Operations Summary

| Start Time | End Time | Dur (hrs) | Phase | Code 1 | Code 2 | Time P-N-T | Comment |
|---|---|---|---|---|---|---|---|
| 00:00 | 00:00 | 24.... | 16" | D | 03 | P | Directional drilling 16" hole section in rotation and slide mode f/ 960m to 1184m w/ 800-850 gpm/ 1850-2050 psi/ 5-20 kips/ 60-70 rpm/ 12-18 kft-lbf.<br><br>Rotation Intervals Slide Interval<br>960m - 984m 984m - 988m<br>988m - 1184m<br><br>** Drilling with NL / NG.<br>** B/R and R/D each stand drilled once.<br>** Pump 20 bbl Hi-Vis pill when stand down.<br>** Take survey before connection.<br>** Max gas @ 1109m = 9.2 %<br><br>>> Found 5 jts 13 3/8" casing with damaged pin. |
| 00:00 | 00:00 | | | | | | Apr 11 , 2014<br><br>00:00 - 06:00 Cont' directional drilling 16'' hole section in rotation mode f/ 1184m to 1240m w/ 870 - 800 gpm/ 2150 - 2000 psi/ 16 - 20 kips/ 70 rpm/ 12 - 18 kft-lbf.<br><br>** Drilling with NL / NG.<br>** B/R and R/D each stand drilled once.<br>** Pump 20 bbl Hi-Vis pill when stand down.<br>** Take survey before connection.<br>** Max gas @ 1189m = 5.8 %<br><br>>> Last Sample @ 1230m consists of : Limestone (100 %) |

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### Mud Information

| Depth (mKB) | Type | Density (lb/gal) | Funnel Viscosity (s/qt) | PV Calc (cp) | YP Calc (lbf/100ft²) | Filtrate (mL/30min) |
|---|---|---|---|---|---|---|
| 1,184.00 | Polymer Mud | 9.10 | 50 | 15.0 | 20.000 | 6.0 |
| **pH** | **Pf (mL/mL)** | **Mf (mL/mL)** | **Pm (mL/mL)** | **Sand (%)** | **Low Gravity Solids (%)** | **High Gravity Solids (%)** |
| 9.8 | | | | 2.0 | | |
| **Vis 3rpm** | **Vis 6rpm** | **Percent Oil (%)** | **Percent Water (%)** | **Solids (%)** | **Chlorides (mg/L)** | **Filter Cake (/32")** |
| 4 | 6 | 0.0 | 93.0 | 7.0 | 27,000.000 | |
| **Gel 10 sec (lbf/100ft²)** | **Gel 10 min (lbf/100ft²)** | **Gel 30 min (lbf/100ft²)** | **Oil Water Ratio** | **T (fl) (°C)** | **Active Mud Volume (bbl)** | **Reserve Mud Volume (bbl)** |
| 4.000 | 6.000 | | 0/100 | 46.0 | 1488.0 | 1515.0 |
| **Cum Mud Added (bbl)** | **Mud Lost to Surface (bbl)** | **Mud Lost to Hole (bbl)** | **Avg Background Gas (%)** | **Avg Drill Gas (%)** | **K** | **n** |
| 1060.0 | 100.0 | 0.0 | 1.35 | | 9.892756 | 0.445 |

### Marine Conditions

| Swell Height (m) | Visibility (km) | Wind Direction (°) | Wind Speed (knots) | Barometric Pressure (mmHg) | Temperature - High (°C) |
|---|---|---|---|---|---|
| 0.10 | 7. | 40 | 1.0 | 1,009.0 | 26.0 |

### Support Vessels

| Vessel Name | Type | Arrival Date | Depart | Note |
|---|---|---|---|---|
| F-GVEF | Chopper | 4/10/2014 11:50 | 4/10/2014 11:57 | Pax in: 6 , Pax out: 6 |
| EP-HBJ | Chopper | 4/10/2014 13:01 | 4/10/2014 13:04 | Pax in: 0 , Pax out: 1 |
| Ahmed-06 | Supply Boat | 4/9/2014 21:30 | 4/10/2014 18:45 | Trans Coil Tubing reel to Osam Condor by DD6 Crane, C/Off & Dep for Kish. |
| Osam Condor | Supply Boat | 4/10/2014 17:05 | 4/10/2014 23:00 | Delivered all DD6 Cargo, Rec Cargo from Ahmed 6 for Kish, Dep for Sahar 2. |
| Swissco | Supply Boat | 4/11/2014 04:20 | 4/11/2014 05:00 | Arrived from Aban 8 Location, Transfrd 1 px with his Cargo, C/Off & Dep for Kish, |

### Mud Additives

| Description | Consumed | Cum Rec | On Loc |
|---|---|---|---|
| AMYLOSE B | 41.0 | 400.0 | 80.0 |
| Barite BB | | 15.0 | 15.0 |
| Bio-Cide | 2.0 | 210.0 | 45.0 |
| Bio-XCD | 13.0 | 414.0 | 149.0 |
| Defoam SLC | 3.0 | 220.0 | 74.0 |
| H2S Scavenger | 5.0 | 120.0 | 58.0 |
| Pac-R | 12.0 | 320.0 | 98.0 |
| Pac-UL | 14.0 | 160.0 | 40.0 |
| Caustic Soda | 15.0 | 657.0 | 137.0 |
| CLAY-BOND L | 22.0 | 44.0 | 22.0 |

### Bulk Amounts

| Supply Item Description | Consumed | Received | Cum On Loc |
|---|---|---|---|
| Barite BB | 0.0 | 15.0 | 15.0 |
| Barite Bulk | 0.0 | 0.0 | 147.0 |
| Cement G | 0.0 | 0.0 | 150.0 |
| Bentonite Bulk | 0.0 | 0.0 | 65.0 |
| Bentonite BB | 0.0 | 0.0 | 21.0 |
| Diesel Fuel | 11.0 | 0.0 | 152.0 |
| Potable Water | 20.0 | 20.0 | 110.0 |
| Drill Water | 80.0 | 0.0 | 620.0 |

### Bit/BHA Information

| BHA No. | Size (in) | Make | Model | Serial Number | Bit Type | IADC Codes | Nozzles (/32") |
|---|---|---|---|---|---|---|---|
| 5 | 16 | Varel | VM516DGXU | 6004481 | PDC | M322 | 13/13/13/13/13/13/13 |
| **TFA (in²)** | **Drilling Time (hrs)** | **Bit Revs** | **Depth In (mKB)** | **Depth Out (mKB)** | **Depth Drilled (m)** | **BHA ROP (m/hr)** | **IADC Bit Dull** |
| 1.04 | 24.90 | 250,215.60 | 878.00 | 1,184.00 | 306.00 | 12.3 | ------- |
| **Flow Rate (min) (gpm)** | **Flow Rate (max) (gpm)** | **SPP (min) (psi)** | **SPP (max) (psi)** | **WOB (min) (1000lbf)** | **WOB (max) (1000lbf)** | **RPM (min) (rpm)** | **RPM (max) (rpm)** |
| 750 | 890 | 1,650.0 | 2,100.0 | 10 | 22 | 111 | 181 |
| **Circulating Time (hrs)** | **Rotating Time (hrs)** | **% Rotating Time (%)** | **% Sliding Time (%)** | **PU HL (max) (1000lbf)** | **Rot HL (max) (1000lbf)** | **SO HL (min) (1000lbf)** | **String Wt (1000lbf)** |
| 32.13 | 24.05 | 96.59 | 3.41 | 308 | 285 | 237 | |

### Drilling Parameters

| Depth Start (mKB) | Depth End (mKB) | Drilling Time (hrs) | Drilling Torque | Flow Rate (gpm) | RPM (rpm) | SPP (psi) | WOB (1000lbf) | Drill Str Wt (1000lbf) |
|---|---|---|---|---|---|---|---|---|
| 960.00 | 984.00 | 1.91 | 15,000.0 | 850 | 60 | 2,050.0 | 18 | 285 |
| 984.00 | 988.00 | 0.85 | | 850 | | 1,950.0 | 10 | 285 |
| 988.00 | 1,154.00 | 12.85 | 16,000.0 | 800 | 65 | 1,850.0 | 18 | 285 |
| 1,154.00 | 1,184.00 | 2.59 | 15,000.0 | 850 | 70 | 2,050.0 | 16 | 285 |

### Drill String Components

| Item Description | OD (in) | ID (in) | Len (m) | Cum Len (m) |
|---|---|---|---|---|
| 15 jts 5 7/8" HWDP | 5 7/8 | 4.250 | 139.60 | 313.30 |
| XOS | 8 | 3.000 | 0.79 | 173.70 |
| 6 jts 8 1/4" DC | 8 1/4 | 2.810 | 56.24 | 172.91 |
| 8" DA Drilling Jar (COR2898AD) | 8 | 3.500 | 5.72 | 116.67 |
| 8 jts 8 1/4" DC | 8 1/4 | 2.810 | 75.34 | 110.95 |
| XOS | 9 1/2 | 3.000 | 1.22 | 35.61 |
| 9 1/2" Float sub with valve | 9 1/2 | 3.000 | 0.78 | 34.39 |
| 9 1/2" NMDC | 9 1/2 | 4.000 | 9.44 | 33.61 |
| 9 1/2" TM-Sub (Mwd and gamma probes inside) | 9 1/2 | | 2.11 | 24.17 |
| 9 1/2" NMDC | 9 1/2 | 4.000 | 9.44 | 22.06 |
| 9 1/2" Restrictor Sub (Jet: 36) | 9 1/2 | 4.300 | 0.92 | 12.62 |
| 14 3/4" String Stabilizer (COR3032AD) | 14 3/4 | 3.000 | 2.16 | 11.70 |

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| Item Description | OD (in) | ID (in) | Len (m) | Cum Len (m) |
|---|---|---|---|---|
| 9 5/8" Motor (BH: 1.25 , Sleeve: 15 3/4") | 9 5/8 | | 9.14 | 9.54 |
| 16" PDC Bit (SN: 6004481) | 16 | | 0.40 | 0.40 |

### Well Control

| No. | Depth (mKB) | Stroke (in) | Strokes (spm) | Pres (psi) | Eff (%) | Q (gpm) |
|---|---|---|---|---|---|---|
| 1 | 1,151.00 | 14.00 | 20 | 100.0 | 97 | 114 |
| 1 | 1,151.00 | 14.00 | 30 | 150.0 | 97 | 170 |
| 1 | 1,151.00 | 14.00 | 40 | 235.0 | 97 | 227 |
| 3 | 1,151.00 | 14.00 | 20 | 110.0 | 97 | 114 |
| 3 | 1,151.00 | 14.00 | 30 | 155.0 | 97 | 170 |
| 3 | 1,151.00 | 14.00 | 40 | 230.0 | 97 | 227 |

| Test Date | Last Casing String Run | Test Type | Depth (mKB) | TVD (mKB) | Applied Surface Pressure (psi) |
|---|---|---|---|---|---|
| 4/8/2014 | 18 5/8" Shoe, 870.00mKB | Formation Integrity | 870.00 | 843.25 | 400.0 |

| Applied Surface Pressure (psi) | Volume Pumped (bbl) | Current MAASP | Kick Tolerance | Dens Fluid (lb/gal) | Leak Off Pressure (psi) |
|---|---|---|---|---|---|
| 400.0 | 8.0 | 400.00 | | 11.48 | 1,650.4 |

### Directional Survey Data

| MD (mKB) | Azm (°) | Incl (°) | TVD (mKB) | NS (m) | EW (m) | VS (m) | DLS (°/30m) | Build (°/30m) |
|---|---|---|---|---|---|---|---|---|
| 989.20 | 247.10 | 22.10 | 953.60 | -106.94 | -193.18 | 220.74 | 0.70 | -0.70 |
| 1,018.02 | 247.10 | 22.30 | 980.29 | -111.18 | -203.22 | 231.60 | 0.21 | 0.21 |
| 1,047.60 | 246.70 | 22.90 | 1,007.59 | -115.64 | -213.67 | 242.93 | 0.63 | 0.61 |
| 1,076.90 | 246.90 | 22.90 | 1,034.59 | -120.13 | -224.15 | 254.30 | 0.08 | 0.00 |
| 1,105.87 | 246.70 | 22.50 | 1,061.31 | -124.53 | -234.43 | 265.44 | 0.42 | -0.41 |
| 1,135.13 | 245.60 | 22.80 | 1,088.31 | -129.09 | -244.73 | 276.69 | 0.53 | 0.31 |
| 1,164.34 | 244.70 | 23.10 | 1,115.21 | -133.88 | -255.07 | 288.06 | 0.47 | 0.31 |
| 1,193.36 | 244.20 | 22.10 | 1,142.00 | -138.69 | -265.13 | 299.21 | 1.05 | -1.03 |
| 1,222.12 | 244.10 | 21.50 | 1,168.71 | -143.34 | -274.74 | 309.89 | 0.63 | -0.63 |

### Geology Information

| Formation Name | Prog Top MD (mKB) | Final Top MD (mKB) | Final Top TVD (mKB) |
|---|---|---|---|
| Kazhdumi | | 1,046.00 | 1,006.12 |
| Upper Dariyan | | 1,091.00 | 1,047.56 |
| Hawar | | 1,154.00 | 1,105.73 |
| Lower Dariyan | | 1,183.00 | 1,132.43 |
| Upper Gadvan | | 1,209.00 | 1,156.52 |

### General Remarks

**Comment**

>>> Requirements: NIL

>>> Sent: NIL

>>> Received:

*** Osam Condor boat from Kish base:

(PFD chemicals: 15 BB barite) + (ABAN cargo: 2 no garbage skip) + (AMOS cement chemicals: 1 pallet Retarder + 1 pallet Defoamer + 1 pallet Spacer + 1 pallet Sodium Silicate) + (AMOS 1 no 13 3/8" Single cement head with related XOS) + (AMOS 1 no 13 3/8" Double cement head with related XOS) + (AMOS 2 no 13 3/8" Circulating swage: BTC to 1502) + (PPL 2 no 13 3/8" Circulating swage: BTC to 5 1/2" FH) + (AL-Ghaith 13 3/8" casing handling tools: 1 no Power unit + 2 no Power tong + 2 no Spider slips, 500 T + 2 no Spider elevator, 500 T + 2 no Side door elevator, 250 T + 2 no Single joint elevator + 2 no Manual casing slips + 2 no Fill up tools + 2 no Safety clamp + 4 no Thread protector + 2 no Stabbing guide + 2 no casing drift + 2 Manual tong + 2 no Dope)